



DR. B. C. ROY ENGINEERING COLLEGE, DURGAPUR

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Week 7-8: Transmission Planning and Reliability

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Comprehensive Class Notes

Prepared by:
Kingsuk Majumdar

Department of Electrical Engineering

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1 Module Overview & Weekly Structure

The main aim of Module 4 is to establish comprehensive understanding of transmission system planning methodologies, composite system reliability assessment, and network optimization techniques. This module integrates generation planning concepts with transmission infrastructure requirements, enabling holistic evaluation of power system adequacy and security.

Definition 1.1 (Module Philosophy): *Transmission planning represents the critical link between generation resources and load centers, requiring sophisticated analysis of network topology, transfer capabilities, and system reliability under various operating conditions and contingencies.*

1.1 Weekly Learning Structure

Table 1: Transmission Planning and Reliability Course Structure - Week 7-8

Lecture	Topic	Duration	Learning Activities
4.1	Transmission Planning Objectives	50 min	Network topology analysis
4.2	Network Reconfiguration Strategies	50 min	Graph theory applications
4.3	System Reliability Indices	50 min	Index calculation workshop
4.4	Load Point Reliability Analysis	50 min	Nodal reliability assessment
4.5	Composite System Modeling	50 min	Integrated G&T analysis
4.6	Data Requirements and Management	50 min	Database design principles
4.7	Transmission Reliability Assessment	50 min	Comprehensive case study

2 Transmission Planning Objectives

Transmission planning ensures adequate and reliable power transfer capability between generation sources and load centers while maintaining system security, economic efficiency, and operational flexibility under diverse operating conditions.

2.1 Transmission System Functions

2.1.1 Primary Functions

Definition 2.1 (Transmission System Role): *The transmission system serves as the backbone infrastructure that:*

- *Interconnects geographically dispersed generation facilities*
- *Delivers bulk power to distribution substations*
- *Enables power exchange between interconnected systems*
- *Provides operational flexibility for load following and contingency management*
- *Facilitates economic dispatch and energy markets*

2.1.2 Planning Horizons

Transmission planning operates across multiple time scales:

Long-term Planning (10-30 years):

- Major corridor identification and right-of-way acquisition
- Strategic network expansion for load growth
- Integration of new generation resources
- Regional interconnection development

Medium-term Planning (5-10 years):

- Detailed project development and engineering
- Capacity reinforcement programs
- Substation expansion planning
- Reliability enhancement initiatives

Short-term Planning (1-5 years):

- Project execution and commissioning
- Operational readiness assessment
- Maintenance scheduling optimization
- Emergency response planning

2.2 Planning Objectives and Criteria

2.2.1 Adequacy Objectives

Definition 2.2 (Transmission Adequacy): *Transmission adequacy refers to the ability of the network to transport sufficient power from generation sources to meet load requirements under normal and contingency conditions.*

Key adequacy metrics:

$$\text{Transfer Capability} = \text{Thermal Limit} \wedge \text{Voltage Limit} \wedge \text{Stability Limit} \quad (1)$$

$$\text{Reserve Margin} = \frac{\text{Available Transfer Capability} - \text{Peak Transfer}}{\text{Peak Transfer}} \quad (2)$$

2.2.2 Security Objectives

Definition 2.3 (N-1 Security Criterion): *The system must withstand the loss of any single transmission element (line, transformer, or bus) without:*

- Load curtailment
- Equipment overloading
- Voltage violations
- System instability

Extended security criteria:

N-1-1 Criterion: System must survive a second contingency following an N-1 event with controlled corrective actions.

N-2 Criterion: System must withstand simultaneous loss of two elements for critical facilities.

2.2.3 Economic Objectives

Transmission planning minimizes total system costs while meeting reliability requirements:

$$\min \{C_{\text{capital}} + C_{\text{operating}} + C_{\text{losses}} + C_{\text{reliability}}\} \quad (3)$$

where:

- C_{capital} = Present value of investment costs
- $C_{\text{operating}}$ = Maintenance and operational expenses
- C_{losses} = Value of transmission losses
- $C_{\text{reliability}}$ = Expected cost of load interruptions

2.3 Planning Constraints and Considerations**2.3.1 Technical Constraints**

Thermal constraints:

$$|I_l| \leq I_l^{\max}, \quad \forall l \in \mathcal{L} \quad (4)$$

where I_l is line current and $I_l^{\max}$ is thermal rating.

Voltage constraints:

$$V_i^{\min} \leq |V_i| \leq V_i^{\max}, \quad \forall i \in \mathcal{N} \quad (5)$$

Stability constraints:

- Steady-state stability margin
- Transient stability limits
- Voltage stability boundaries
- Small-signal stability requirements

2.3.2 Environmental and Social Constraints

- Right-of-way restrictions
- Environmental impact assessments
- Land acquisition challenges
- Public acceptance and aesthetic concerns
- Electromagnetic field (EMF) regulations
- Wildlife and habitat protection

Example 2.1 (Planning Objective Application):

Consider a regional transmission system with:

- Peak load: 5000 MW
- Import capability: 1200 MW
- Local generation capacity: 4500 MW
- N-1 transfer limit: 1000 MW

Adequacy assessment:

$$\text{Required import} = 5000 - 4500 = 500 \text{ MW} \quad (6)$$

$$\text{N-1 capability} = 1000 \text{ MW} > 500 \text{ MW} \quad \checkmark \quad (7)$$

Security margin:

$$\text{Margin} = \frac{1000 - 500}{500} = 1.0 = 100\% \quad (8)$$

System meets adequacy requirements with substantial security margin.

3 Network Reconfiguration Strategies

Network reconfiguration optimizes transmission system topology to enhance reliability, reduce losses, improve voltage profiles, and increase transfer capabilities through strategic switching operations and infrastructure modifications.

3.1 Graph Theory Applications

3.1.1 Network Representation

Definition 3.1 (Transmission Network Graph): A transmission network is represented as a graph $G = (\mathcal{N}, \mathcal{E})$ where:

- $\mathcal{N}$ = Set of nodes (buses/substations)
- $\mathcal{E}$ = Set of edges (transmission lines/transformers)
- $n = |\mathcal{N}|$ = Number of nodes
- $m = |\mathcal{E}|$ = Number of edges

3.1.2 Network Topology Metrics

Connectivity measures:

Node degree:

$$d_i = \sum_{j \in \mathcal{N}} a_{ij} \quad (9)$$

where a_{ij} is the adjacency matrix element.

Network density:

$$\rho = \frac{2m}{n(n-1)} \quad (10)$$

Average path length:

$$L = \frac{1}{n(n-1)} \sum_{i \neq j} d_{ij} \quad (11)$$

where d_{ij} is shortest path length between nodes i and j .

3.2 Reconfiguration Objectives

3.2.1 Reliability Enhancement

Redundancy improvement:

Creating parallel paths increases system reliability:

$$R_{\text{parallel}} = 1 - \prod_{i=1}^n (1 - R_i) \quad (12)$$

Critical element identification:

Elements whose removal most degrades network connectivity are prioritized for redundancy:

$$\text{Criticality}_l = \Delta\rho_l = \rho - \rho_{-l} \quad (13)$$

where ρ_{-l} is density after removing element l .

3.2.2 Loss Reduction

Network reconfiguration can reduce transmission losses:

$$\min \sum_{l \in \mathcal{E}} R_l |I_l|^2 \quad (14)$$

subject to:

- Power flow equations
- Radial or meshed topology constraints
- Voltage and thermal limits

3.2.3 Transfer Capability Enhancement

Strategic reconfiguration increases Available Transfer Capability (ATC):

$$\text{ATC} = \text{TTC} - \text{TRM} - \text{ETC} - \text{CBM} \quad (15)$$

$$= \text{TTC} - \text{Margins} \quad (16)$$

where:

- TTC = Total Transfer Capability
- TRM = Transmission Reliability Margin
- ETC = Existing Transmission Commitments
- CBM = Capacity Benefit Margin

Example 3.1 (Network Reconfiguration Analysis):

Consider a 6-bus transmission system with the following characteristics:

Initial configuration:

- Total losses: 12.5 MW
- Minimum voltage: 0.92 p.u.

- Maximum line loading: 85%

Available switching actions:

1. Open line 2-3, close line 2-4
2. Open line 4-5, close line 3-5
3. Open line 1-3, close line 1-4

Post-reconfiguration analysis:

Action 1 results:

- Losses: 11.8 MW (5.6% reduction)
- Minimum voltage: 0.94 p.u. (improved)
- Maximum loading: 78% (improved)
- N-1 security: Maintained

Recommendation: Implement Action 1 for optimal performance improvement.

4 System Reliability Indices

System reliability indices quantify transmission network performance, enabling comparison of design alternatives, tracking performance trends, and establishing reliability targets for planning and operation.

4.1 Hierarchical Reliability Indices

4.1.1 Hierarchical Level II (HL-II) Indices

Definition 4.1 (Composite System Reliability): *HL-II indices evaluate combined generation and transmission system reliability, considering both resource adequacy and network capability to deliver power to load points.*

System indices:

Loss of Load Probability (LOLP):

$$\text{LOLP} = \sum_{s \in \mathcal{S}_{\text{fail}}} P(s) \quad (17)$$

where $\mathcal{S}_{\text{fail}}$ is the set of system states with load curtailment.

Expected Energy Not Supplied (EENS):

$$\text{EENS} = \sum_{s \in \mathcal{S}} P(s) \cdot E_s \cdot T \quad (18)$$

where E_s is energy curtailment in state s and T is time period.

Loss of Load Expectation (LOLE):

$$\text{LOLE} = \sum_{t=1}^T \text{LOLP}_t \cdot \Delta t \quad (19)$$

Standard criterion: $\text{LOLE} \leq 0.1$ days/year

4.1.2 Load Point Indices

Probability of Load Curtailment (PLC):

$$PLC_i = \sum_{s \in \mathcal{S}_i} P(s) \quad (20)$$

where $\mathcal{S}_i$ is the set of states causing curtailment at load point i .

Expected Demand Not Supplied (EDNS):

$$EDNS_i = \sum_{s \in \mathcal{S}_i} P(s) \cdot C_{s,i} \quad (21)$$

where $C_{s,i}$ is curtailment at point i in state s .

Frequency of Load Curtailment:

$$f_i = \sum_{s \in \mathcal{S}_i} f_s \quad (22)$$

where f_s is frequency of entering state s .

Average Duration of Curtailment:

$$D_i = \frac{PLC_i}{f_i} \quad (23)$$

4.2 Network Performance Indices

4.2.1 Transfer Capability Indices

Available Transfer Capability (ATC):

Maximum additional power transfer possible over interconnection:

$$ATC = \min \{ ATC_{\text{thermal}}, ATC_{\text{voltage}}, ATC_{\text{stability}} \} \quad (24)$$

Transmission Reliability Margin (TRM):

Uncertainty allowance in transfer capability calculation:

$$TRM = k \cdot \sigma_{TTC} \quad (25)$$

where k is confidence level factor (typically 1.96 for 95% confidence) and σ_{TTC} is standard deviation of TTC.

4.2.2 Voltage Performance Indices

Voltage Deviation Index:

$$VDI = \frac{1}{n} \sum_{i=1}^n \left| \frac{V_i - V_i^{\text{nom}}}{V_i^{\text{nom}}} \right| \quad (26)$$

Voltage Stability Margin:

$$VSM = \frac{P_{\text{critical}} - P_{\text{operating}}}{P_{\text{operating}}} \times 100\% \quad (27)$$

where P_{critical} is loading level at voltage collapse point.

4.3 Economic Performance Indices

4.3.1 Cost-Reliability Indices

Interrupted Energy Assessment Rate (IEAR):

$$\text{IEAR} = \frac{\text{Total Customer Interruption Cost}}{\text{EENS}} \quad (28)$$

Typical range: 5-50 USD/kWh

System Interruption Cost:

$$C_{\text{int}} = \sum_{i=1}^n \text{EDNS}_i \cdot \text{IEAR}_i \quad (29)$$

Cost-Benefit Ratio:

$$\text{BCR} = \frac{\text{PV}(\Delta C_{\text{int}})}{\text{PV}(C_{\text{invest}} + C_{\text{O\&M}})} \quad (30)$$

where PV denotes present value.

Example 4.1 (System Reliability Index Calculation):

Consider a composite generation-transmission system with the following annual data:

System characteristics:

- Peak load: 3500 MW
- Number of load points: 50
- Total energy served: 25,000 GWh
- Average IEAR: \$15/kWh

Reliability performance:

- Total hours with curtailment: 2.5 hours
- Total energy curtailed: 450 MWh
- Number of curtailment events: 5

Calculate indices:

LOLE:

$$\text{LOLE} = \frac{2.5}{24} = 0.104 \text{ days/year} \quad (31)$$

Slightly exceeds 0.1 day/year criterion.

EENS:

$$\text{EENS} = 450 \text{ MWh/year} \quad (32)$$

Customer interruption cost:

$$C_{\text{int}} = 450,000 \times 15 = \$6,750,000/\text{year} \quad (33)$$

Assessment: System performance marginally below target. Consider transmission reinforcement if cost-effective ($\text{BCR} > 1$).

5 Load Point Reliability Analysis

Load point reliability analysis evaluates power supply adequacy at individual delivery points, accounting for generation resource availability, transmission path reliability, and substation equipment performance.

5.1 Load Point Evaluation Methods

5.1.1 Minimal Cut Set Method

Definition 5.1 (Minimal Cut Set): A minimal cut set is the smallest collection of components whose simultaneous failure causes load point interruption. Removing any component from the set restores connectivity.

Load point unavailability:

$$U_i = \sum_{k=1}^K U_{C_k} \quad (34)$$

where U_{C_k} is unavailability of cut set k :

$$U_{C_k} = \prod_{j \in C_k} U_j \quad (35)$$

For independent components with small unavailabilities, second-order terms are often negligible:

$$U_i \approx \sum_{k=1}^K U_{C_k} \quad (36)$$

5.1.2 Conditional Probability Approach

Event tree analysis:

Systematically enumerate all component state combinations leading to load point failure:

$$PLC_i = \sum_{s \in \mathcal{S}_{\text{fail}}} \prod_{j \in \mathcal{U}_s} U_j \prod_{k \in \mathcal{A}_s} A_k \quad (37)$$

where $\mathcal{U}_s$ and $\mathcal{A}_s$ are sets of unavailable and available components in state s .

5.2 Multi-State System Analysis

5.2.1 Capacity State Enumeration

For systems with derated states:

$$PLC_i = \sum_s P_s \cdot I_s \quad (38)$$

where:

$$I_s = \begin{cases} 1 & \text{if capacity} < \text{demand in state } s \\ 0 & \text{otherwise} \end{cases} \quad (39)$$

Partial curtailment:

$$EDNS_i = \sum_s P_s \cdot \max(0, D_i - C_s) \quad (40)$$

where C_s is available capacity in state s .

5.3 Reliability Worth Assessment

5.3.1 Customer Cost Functions

Sector-specific interruption costs:

Customer Class	IEAR (USD/kWh)	Fixed Cost (USD)
Residential	5-15	0.5-2
Commercial	15-50	2-10
Industrial	50-200	10-100

Duration-dependent costs:

$$C_i(d) = C_{i,0} + c_{i,1}d + c_{i,2}d^2 \quad (41)$$

Example 5.1 (Load Point Reliability Assessment):

Consider load point LP-5 supplied through the following configuration:

Supply paths:

- Path 1: Line A (100 MW) → Transformer T1 (80 MVA)
- Path 2: Line B (100 MW) → Transformer T2 (80 MVA)
- Load: 120 MW peak

Component reliability data:

Component	λ (fail/yr)	r (hrs)
Line A	0.8	12
Line B	1.2	10
Transformer T1	0.1	48
Transformer T2	0.1	48

Component unavailabilities:

$$U_A = \frac{0.8 \times 12}{8760} = 0.00109 \quad (42)$$

$$U_B = \frac{1.2 \times 10}{8760} = 0.00137 \quad (43)$$

$$U_{T1} = \frac{0.1 \times 48}{8760} = 0.00055 \quad (44)$$

$$U_{T2} = \frac{0.1 \times 48}{8760} = 0.00055 \quad (45)$$

Load point unavailability:

For dual path system, both paths must fail:

$$U_{LP} \approx (U_A + U_{T1}) \times (U_B + U_{T2}) = 0.00164 \times 0.00192 = 3.15 \times 10^{-6} \quad (46)$$

Annual outage time:

$$\text{Outage} = 3.15 \times 10^{-6} \times 8760 = 0.028 \text{ hours/year} \approx 1.7 \text{ min/year} \quad (47)$$

Excellent reliability due to redundant supply paths.

6 Composite System Modeling

Composite system models integrate generation adequacy and transmission network capability, enabling comprehensive reliability assessment that accounts for both resource availability and delivery infrastructure.

6.1 Integrated G&T Modeling Framework

6.1.1 System State Space

Definition 6.1 (Composite System State): A composite system state s is characterized by:

$$s = (G, T, L) \quad (48)$$

where:

- G = Generation capacity vector
- T = Transmission element status vector
- L = Load level vector

State space size:

For n_g generators and n_t transmission elements:

$$|\mathcal{S}| = 2^{n_g} \times 2^{n_t} \times n_l \quad (49)$$

where n_l is number of load levels considered.

6.1.2 State Probability Calculation

Assuming independence between generation and transmission:

$$P_s = P_G \times P_T \times P_L \quad (50)$$

Generation state probability:

$$P_G = \prod_{i \in \mathcal{G}_{\text{up}}} A_i \prod_{j \in \mathcal{G}_{\text{down}}} U_j \quad (51)$$

Transmission state probability:

$$P_T = \prod_{k \in \mathcal{T}_{\text{up}}} A_k \prod_{l \in \mathcal{T}_{\text{down}}} U_l \quad (52)$$

6.2 State Evaluation Methods

6.2.1 Optimal Power Flow Analysis

For each system state, solve:

$$\min \sum_i C_i P_{G,i} + \sum_j C_{\text{curt}} P_{\text{curt},j} \quad (53)$$

$$\text{s.t.} \quad \sum_i P_{G,i} - \sum_j P_{D,j} + \sum_j P_{\text{curt},j} - P_{\text{loss}} = 0 \quad (54)$$

$$P_{G,i}^{\min} \leq P_{G,i} \leq P_{G,i}^{\max}, \quad \forall i \quad (55)$$

$$-P_l^{\max} \leq P_l \leq P_l^{\max}, \quad \forall l \quad (56)$$

$$V_i^{\min} \leq |V_i| \leq V_i^{\max}, \quad \forall i \quad (57)$$

6.2.2 DC Load Flow Approximation

Simplified linear model for large-scale studies:

$$P_l = \frac{1}{X_l}(\theta_i - \theta_j) \quad (58)$$

Power balance:

$$\mathbf{B}\boldsymbol{\theta} = \mathbf{P}_{\text{net}} \quad (59)$$

where $\mathbf{B}$ is susceptance matrix.

6.3 Contingency Selection and Ranking

6.3.1 Contingency Screening

Probability threshold:

Analyze states with probability exceeding threshold:

$$P_s \geq P_{\text{threshold}} \quad (60)$$

Typical threshold: 10^{-6} to 10^{-8}

Impact-based screening:

Priority score:

$$S_c = P_c \times I_c \quad (61)$$

where I_c is severity measure (MW curtailed, overloads, voltage violations).

6.3.2 Common Mode Failures

Account for dependent outages:

$$P_{i,j}^{\text{common}} = P_{\text{event}} \times P_{i|\text{event}} \times P_{j|\text{event}} \quad (62)$$

Common causes:

- Weather events (storms, ice, lightning)
- Cascading failures
- Protection system misoperation
- Human error

Example 6.1 (Composite System Reliability Evaluation):

Evaluate reliability of a 5-bus system with 3 generators and 6 transmission lines.

System data:

- Peak load: 400 MW (distributed across buses 3-5)
- Generator capacities: G1=200 MW, G2=150 MW, G3=100 MW
- Generator FORs: 0.05, 0.06, 0.08
- Transmission line failure rates: 0.5-1.5 fail/yr
- Average repair time: 10-20 hours

State space analysis:

$$\text{Total states} = 2^3 \times 2^6 = 512 \text{ combinations} \quad (63)$$

Significant state selection:

Using probability threshold of 10^{-6} :

- Single contingencies: 9 states
- Double contingencies: 36 states
- Higher order: 3 critical states
- Total analyzed: 48 states (9.4% of state space)

Aggregated results:

$$\text{LOLP} = 0.00234 \quad (64)$$

$$\text{LOLE} = 0.00234 \times 365 = 0.854 \text{ days/year} \quad (65)$$

$$\text{EENS} = 1250 \text{ MWh/year} \quad (66)$$

Assessment: System does not meet 0.1 day/year LOLE criterion. Recommend transmission reinforcement.

7 Data Requirements and Management

Comprehensive and accurate data forms the foundation of transmission reliability assessment. Systematic data collection, validation, and management ensure meaningful analysis and informed planning decisions.

7.1 Essential Data Categories

7.1.1 Network Topology Data

Bus/node data:

- Bus identification and naming
- Voltage level and base values
- Geographic coordinates
- Load characteristics
- Generation connections

Branch/element data:

- From/to bus connectivity
- Impedance parameters (R, X, B)
- Thermal ratings (normal, emergency, seasonal)
- Transformer tap settings
- Switch/breaker status

7.1.2 Reliability Parameter Data

Component reliability statistics:

Component	Failure Rate	Repair Time
Overhead line (per 100 km)	0.5-2.0 fail/yr	4-12 hrs
Underground cable (per km)	0.01-0.05 fail/yr	100-500 hrs
Circuit breaker	0.002-0.01 fail/yr	10-50 hrs
Power transformer	0.005-0.02 fail/yr	200-1000 hrs
Busbar	0.001-0.005 fail/yr	50-200 hrs

Failure mode data:

- Permanent vs. temporary failures
- Weather-related failure rates
- Age-dependent failure characteristics
- Common cause failure probabilities

7.2 Data Validation and Quality Control

7.2.1 Consistency Checks

Topology validation:

- Connectivity verification
- Radial/meshed structure confirmation
- Island detection
- Reference bus identification

Parameter range checks:

$$x_{\min} \leq x_i \leq x_{\max}, \quad \forall i \quad (67)$$

Power balance verification:

$$\left| \sum P_G - \sum P_D - P_{\text{loss}} \right| < \epsilon_P \quad (68)$$

7.2.2 Statistical Validation

Outlier detection:

Identify anomalous values using z-score:

$$z_i = \frac{x_i - \mu}{\sigma} \quad (69)$$

Flag if $|z_i| > 3$.

Confidence intervals:

For reliability parameters:

$$\lambda \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \quad (70)$$

7.3 Data Uncertainty Management

7.3.1 Parameter Uncertainty Quantification

Probabilistic representation:

Model uncertainty using distributions:

$$\lambda \sim \mathcal{N}(\mu_\lambda, \sigma_\lambda^2) \quad (71)$$

Sensitivity analysis:

Evaluate impact of parameter variations:

$$\frac{\partial \text{Index}}{\partial \theta} \approx \frac{\text{Index}(\theta + \Delta\theta) - \text{Index}(\theta)}{\Delta\theta} \quad (72)$$

8 Transmission Reliability Assessment

Comprehensive transmission reliability assessment integrates network analysis, contingency evaluation, and performance metrics to provide holistic understanding of system adequacy and identify improvement opportunities.

8.1 Assessment Methodology Framework

8.1.1 Sequential Assessment Process

Phase 1: System characterization

1. Collect network topology and parameter data
2. Establish reliability models for components
3. Define operating scenarios and load levels
4. Identify critical load points

Phase 2: Contingency analysis

1. Enumerate system states
2. Screen and rank contingencies
3. Perform power flow analysis
4. Identify constraint violations

Phase 3: Reliability evaluation

1. Calculate system-level indices
2. Compute load point indices
3. Assess economic impacts
4. Determine weak areas

Phase 4: Enhancement planning

1. Identify improvement alternatives
2. Evaluate cost-effectiveness
3. Rank investment priorities
4. Develop implementation plan

8.2 Analytical Methods

8.2.1 Deterministic Analysis

N-1 security assessment:

For each single contingency c :

$$\text{Solve: } \min f(\mathbf{x}) \quad (73)$$

$$\text{s.t. } g(\mathbf{x}) = \mathbf{0} \quad (74)$$

$$h_c(\mathbf{x}) \leq \mathbf{0} \quad (75)$$

Severity ranking:

$$S_c = w_1 \sum_l \text{OL}_l + w_2 \sum_i \text{VV}_i + w_3 \sum_j \text{LC}_j \quad (76)$$

where OL = overloads, VV = voltage violations, LC = load curtailments.

8.2.2 Probabilistic Analysis

Enumeration method:

1. Generate all significant system states
2. For each state s :
 - Calculate state probability P_s
 - Solve AC/DC power flow
 - Determine curtailments C_s
 - Weight by probability: $P_s \times C_s$
3. Aggregate weighted curtailments

8.3 Reliability Improvement Strategies

8.3.1 Network Reinforcement

Capacity additions:

- New transmission line construction
- Conductor upgrading (reconductoring)
- Series compensation
- Transformer capacity increase

Benefit assessment:

$$\Delta \text{Benefit} = (\text{EENS}_{\text{before}} - \text{EENS}_{\text{after}}) \times \text{IEAR} \quad (77)$$

8.3.2 Technology Integration

FACTS devices:

- SVC (Static VAR Compensator)
- STATCOM (Static Synchronous Compensator)
- TCSC (Thyristor Controlled Series Capacitor)
- UPFC (Unified Power Flow Controller)

Impact on reliability:

- Increased transfer capability
- Enhanced voltage control
- Improved stability margins
- Dynamic congestion management

9 Practice Problems

9.1 Transmission Planning Fundamentals

Problem 9.1 (N-1 Security Assessment):

A transmission corridor consists of two 400 kV lines, each rated 1000 MW. The corridor normally transfers 1400 MW. Determine if the system meets N-1 security criterion.

Solution:

Normal condition:

- Total capacity: $2 \times 1000 = 2000$ MW
- Transfer: 1400 MW
- Loading: $\frac{1400}{2000} = 0.70 = 70\%$

N-1 condition (one line out):

- Remaining capacity: 1000 MW
- Required transfer: 1400 MW
- Deficit: $1400 - 1000 = 400$ MW

Conclusion: System violates N-1 criterion by 400 MW. Requires load curtailment or generation redispatch under single contingency.

Required capacity: To meet N-1 with no curtailment:

$$C_{\text{required}} = 2 \times 1400 = 2800 \text{ MW total capacity} \quad (78)$$

Need additional 800 MW capacity (e.g., third line or higher-capacity conductors).

9.2 System Reliability Indices

Problem 9.2 (System LOLP Calculation):

A composite system has the following contingency analysis results:

State	Probability	Load Curtailment
1	0.00450	0 MW
2	0.00085	120 MW
3	0.00032	250 MW
4	0.00018	180 MW
5	0.00009	350 MW

Calculate LOLP and LOLE.

Solution:

LOLP calculation:

$$\text{LOLP} = P_2 + P_3 + P_4 + P_5 \quad (79)$$

$$= 0.00085 + 0.00032 + 0.00018 + 0.00009 \quad (80)$$

$$= 0.00144 = 0.144\% \quad (81)$$

LOLE calculation:

$$\text{LOLE} = 0.00144 \times 365 = 0.526 \text{ days/year} \quad (82)$$

Assessment: LOLE significantly exceeds 0.1 day/year criterion. System requires reliability enhancement.

EENS calculation:

$$\text{EENS} = \sum P_i \cdot C_i \cdot 8760 \quad (83)$$

$$= (0.00085 \times 120 + 0.00032 \times 250 \quad (84)$$

$$+ 0.00018 \times 180 + 0.00009 \times 350) \times 8760 \quad (85)$$

$$= 2,154 \text{ MWh/year} \quad (86)$$

9.3 Economic Evaluation

Problem 9.3 (Economic Justification):

Proposed transmission reinforcement costs \$80M and will reduce EENS from 4500 MWh/year to 1200 MWh/year. IEAR is \$18/kWh. Annual O&M cost is \$1.2M. Project life is 30 years, discount rate is 6%. Calculate NPV and BCR.

Solution:

Annual benefits:

$$\Delta \text{EENS} = 4500 - 1200 = 3300 \text{ MWh/year} \quad (87)$$

$$B_{\text{annual}} = 3,300,000 \times 18 = \$59,400,000/\text{year} \quad (88)$$

Present value factor:

$$\text{PVF} = \frac{1 - (1.06)^{-30}}{0.06} = 13.765 \quad (89)$$

Present value of benefits:

$$PV_{\text{benefits}} = 59.4 \times 13.765 = \$817.6\text{M} \quad (90)$$

Present value of O&M:

$$PV_{\text{O\&M}} = 1.2 \times 13.765 = \$16.5\text{M} \quad (91)$$

Total costs:

$$PV_{\text{costs}} = 80 + 16.5 = \$96.5\text{M} \quad (92)$$

NPV:

$$NPV = 817.6 - 96.5 = \$721.1\text{M} \quad (93)$$

BCR:

$$BCR = \frac{817.6}{96.5} = 8.47 \quad (94)$$

Conclusion: Highly economically justified ($BCR \gg 1$, large positive NPV).

10 Summary and Key Learning Outcomes

10.1 Fundamental Concepts Mastered

Through comprehensive coverage of Module 4, students have developed thorough understanding of:

1. **Transmission planning fundamentals:** Objectives, constraints, criteria, and multi-horizon planning methodologies
2. **Network optimization:** Reconfiguration strategies, graph theory applications, and topology enhancement techniques
3. **Reliability indices:** System-level, load point, and network performance metrics for comprehensive assessment
4. **Load point analysis:** Minimal cut set methods, composite evaluation, and economic worth assessment
5. **Composite modeling:** Integrated generation-transmission reliability evaluation frameworks
6. **Data management:** Collection, validation, database design, and uncertainty quantification
7. **Assessment methodology:** Deterministic and probabilistic evaluation, benchmarking, and improvement planning

10.2 Practical Applications

The theoretical foundations enable practical applications in:

- **Transmission planning:** Infrastructure expansion, capacity reinforcement, and strategic development
- **Reliability assessment:** System adequacy evaluation, weak area identification, and performance tracking

- **Investment justification:** Cost-benefit analysis, alternative comparison, and project prioritization
- **Operational planning:** Contingency preparation, maintenance scheduling, and emergency response
- **Regulatory compliance:** Meeting reliability standards, documenting performance, and stakeholder reporting
- **Technology evaluation:** FACTS integration, protection enhancement, and smart grid implementation

10.3 Key Mathematical Relationships

Essential transmission reliability equations:

$$\text{ATC: } \text{ATC} = \text{TTC} - \text{TRM} - \text{ETC} - \text{CBM} \quad (95)$$

$$\text{Load Point Unavailability: } U_i = \sum_{k=1}^K \prod_{j \in C_k} U_j \quad (96)$$

$$\text{LOLP: } \text{LOLP} = \sum_{s \in \mathcal{S}_{\text{fail}}} P_s \quad (97)$$

$$\text{EENS: } \text{EENS} = \sum_s P_s \cdot T_s \cdot \sum_i C_{s,i} \quad (98)$$

$$\text{BCR: } \text{BCR} = \frac{\text{PV}(\text{Benefits})}{\text{PV}(\text{Costs})} \quad (99)$$

Definition 10.1 (Module Integration): Module 4 bridges generation planning (Module 3) with distribution reliability (Module 5), providing the critical transmission network perspective that enables comprehensive power system adequacy assessment across all hierarchical levels.

10.4 Assessment and Evaluation

Module 4 assessment components:

- **Class Discussion (15%):** Network topology analysis and reliability concepts
- **Programming Assignment (20%):** Transmission reliability index calculation and composite system evaluation
- **Class Test (5%):** Comprehensive examination of transmission planning theory

Programming assignment deliverables:

1. Network topology analyzer with graph theory metrics
2. Load point reliability calculator using minimal cut set method
3. Composite system reliability evaluator with OPF integration
4. Contingency ranking and screening algorithm
5. Economic assessment module with cost-benefit analysis
6. Comprehensive documentation with test cases and validation

10.5 Next Module Preview

Module 5 will build upon transmission reliability foundations to explore:

- **Distribution system planning:** Network configuration, expansion strategies, and reliability optimization
- **Radial network analysis:** Series components, protection coordination, and restoration procedures
- **Customer reliability indices:** SAIFI, SAIDI, CAIDI, and their calculation methodologies
- **Distributed generation:** Integration impacts, reliability benefits, and operational challenges
- **Smart distribution:** Automation, self-healing networks, and advanced technologies
- **Reliability-centered maintenance:** Predictive strategies and asset management optimization

11 Nomenclature and Abbreviations

11.1 Symbols

Symbol	Description
λ	Failure rate (failures/year)
μ	Repair rate (repairs/year)
U	Unavailability
A	Availability
R	Reliability
P_s	State probability
C_s	Curtailement in state s (MW)
f	Frequency (events/year)
D	Duration (hours)
ρ	Network density
$\mathcal{N}$	Set of nodes
$\mathcal{E}$	Set of edges
I_l	Line current
V_i	Bus voltage
P_G	Generation power
P_D	Demand power

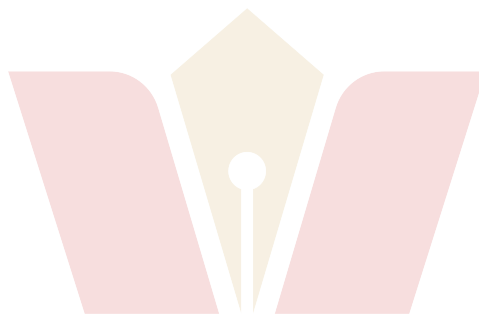
11.2 Abbreviations

Abbreviation	Full Form
ATC	Available Transfer Capability
BCR	Benefit-Cost Ratio
CBM	Capacity Benefit Margin
EDNS	Expected Demand Not Supplied
EENS	Expected Energy Not Supplied
ETC	Existing Transmission Commitments
FACTS	Flexible AC Transmission Systems
FOR	Forced Outage Rate
G&T	Generation and Transmission
HL-II	Hierarchical Level II
IEAR	Interrupted Energy Assessment Rate
LOLE	Loss of Load Expectation
LOLP	Loss of Load Probability
MTBF	Mean Time Between Failures
MTTF	Mean Time To Failure
MTTR	Mean Time To Repair
NPV	Net Present Value
OPF	Optimal Power Flow
O&M	Operation and Maintenance
PLC	Probability of Load Curtailment
PV	Present Value
STATCOM	Static Synchronous Compensator
SVC	Static VAR Compensator
TCSC	Thyristor Controlled Series Capacitor
TRM	Transmission Reliability Margin
TTC	Total Transfer Capability
UPFC	Unified Power Flow Controller

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End of Module 4 Class Notes

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Document Prepared by: Kingsuk Majumdar, Ph.D (Engg)
Department of Electrical Engineering
Dr. B. C. Roy Engineering College, Durgapur

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